ML Optimisation Progress Documentation

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Model Performance on Training Data

Model Testing 1 (21/2/26)

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--- Core performance metrics ---

[Test 1] Accuracy: 79.34%

[Test 2] Weighted F1-score: 79.27%

[Test 3] Classification Report:
              precision    recall  f1-score   support

     Comment       1.00      0.94      0.97        17
        Like       0.60      0.61      0.61        49
        Play       0.90      0.94      0.92        50
      Search       1.00      1.00      1.00        47
   Subscribe       0.60      0.58      0.59        50

    accuracy                           0.79       213
   macro avg       0.82      0.81      0.82       213
weighted avg       0.79      0.79      0.79       213

[Test 4] Confusion Matrix:
[[16  0  1  0  0]
 [ 0 30  2  0 17]
 [ 0  1 47  0  2]
 [ 0  0  0 47  0]
 [ 0 19  2  0 29]]

[Test 5] ROC-AUC Score:
ROC-AUC Score (multi-class, OVR): 0.9557

[Test 6] Per-class accuracy:
Comment: 94.12%
Like: 61.22%
Play: 94.00%
Search: 100.00%
Subscribe: 58.00%
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[Test 8] Permutation Importance:
duration              2.638498e-01
std_iat               5.164319e-02
avg_outbound_size     1.408451e-02
avg_iat               7.042254e-03
pk_count              2.816901e-03
avg_inbound_size      1.110223e-17
total_bytes          -1.408451e-03
outbound_ratio       -4.694836e-03
std_pkt_size         -5.164319e-03
avg_pkt_size         -5.164319e-03
max_pkt_size         -7.981221e-03
dtype: float64

[Test 9] Feature ablation results:
Baseline accuracy: 79.34%
Without pk_count: 79.34%
Without total_bytes: 75.12%
Without avg_pkt_size: 78.40%
Without std_pkt_size: 76.53%
Without max_pkt_size: 78.40%
Without avg_iat: 78.87%
Without std_iat: 77.46%
Without duration: 77.93%
```

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--- Model robustness & reliability ---

[Test 10] Overfitting check:

Training Accuracy: 100.00%
Testing Accuracy:  79.34%
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[Test 11] Class distribution:
Total samples: 645
action
Like         150
Play         150
Subscribe    150
Search       143
Comment       52
Name: count, dtype: int64

[Test 12]Prediction confidence (first 10 rows):
Sample 0: 0.910
Sample 1: 0.605
Sample 2: 0.970
Sample 3: 0.940
Sample 4: 0.670
Sample 5: 1.000
Sample 6: 0.495
Sample 7: 0.600
Sample 8: 0.835
Sample 9: 1.000

[Test 13] Noise sensitivity test:
Noise 0.5% -> Accuracy: 81.22%
Noise 1.0% -> Accuracy: 77.46%
Noise 5.0% -> Accuracy: 78.87%
```

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[Test 14] Stability test:

Stability scores: [0.8169014084507042, 0.82629
10798122066, 0.8262910798122066, 0.81690140845
07042, 0.7840375586854460]
Mean stability: 0.8140845070422535
```

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--- Error & unseen label analysis ---

[Test 15] Error analysis:

Misclassified samples:
     pk_count  total_bytes  ...       true       pred
380      2079      2275493  ...  Subscribe       Like
86       4981      5571646  ...       Like  Subscribe
429      2322      2512597  ...  Subscribe       Like
372      1839      1949806  ...  Subscribe       Like
129      3728      4117888  ...       Like  Subscribe
416      2576      2786165  ...  Subscribe       Like
395      5168      5824475  ...  Subscribe       Like
432      6049      6915035  ...  Subscribe       Like
83       5571      6361361  ...       Like  Subscribe
366      7274      8205666  ...  Subscribe       Like
402      2084      2161600  ...  Subscribe       Like
220      1572      6319154  ...       Play  Subscribe
373      3021      3204798  ...  Subscribe       Like
61       4129      4476640  ...       Like       Play
87       2644      2916359  ...       Like  Subscribe
8         872      1900734  ...       Like  Subscribe
149      1465      1552587  ...       Like  Subscribe
110      2866      3123472  ...       Like  Subscribe
423      6349      7213319  ...  Subscribe       Like
70       1727      1775297  ...       Like  Subscribe

[20 rows x 13 columns]
Total misclassified: 44

[Test 16] Testing model handling of unseen labels:

Unknown label handling test passed.
Model correctly raises error for unseen class:
Error: y contains previously unseen labels: 'new_action'

[test 17] General Sanity check (first 5 rows of features):
     pk_count  total_bytes  ...  avg_outbound_size  avg_inbound_size
485      7342      8039065  ...         235.863911       1229.147717
449      4782      5405458  ...         188.973638       1257.520057
460     10287     11589139  ...         158.495955       1258.782234
488      4328      4424436  ...         332.895161       1165.391183
345       741      1907284  ...         634.190476       3573.554192

[5 rows x 11 columns]
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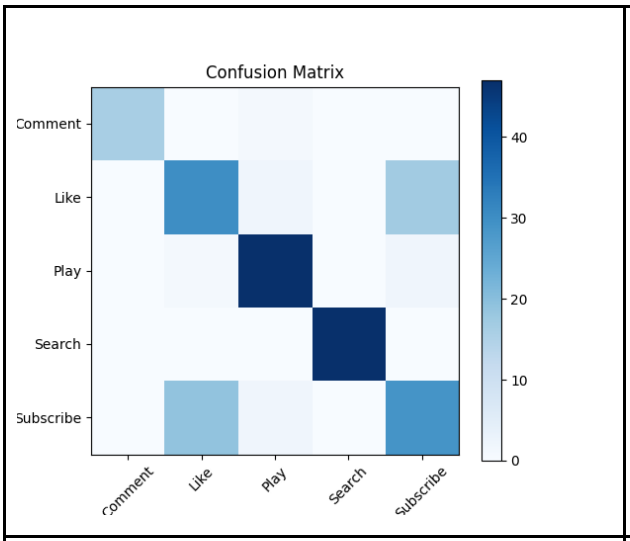

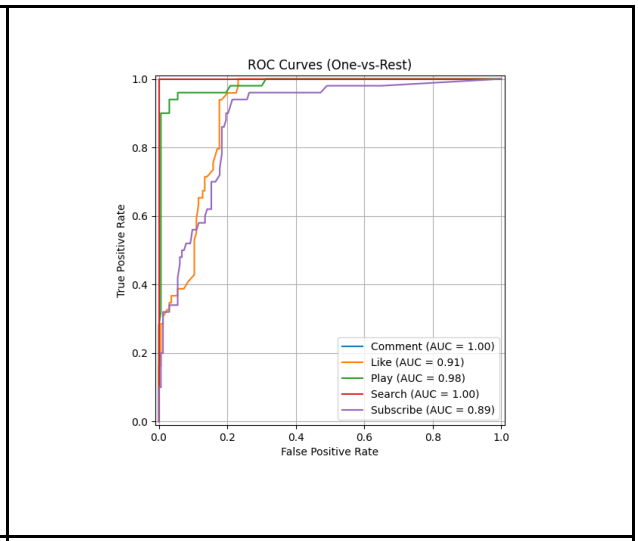

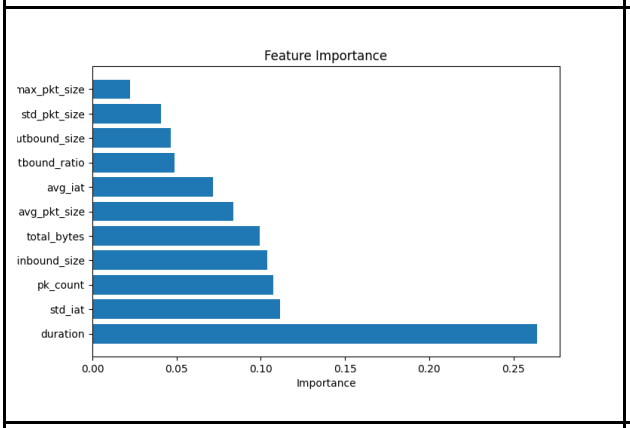

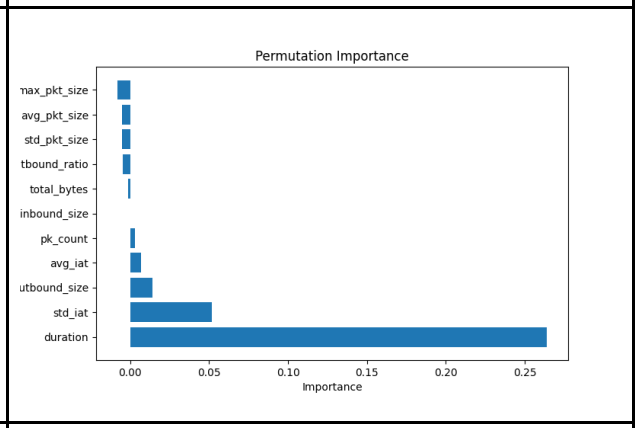

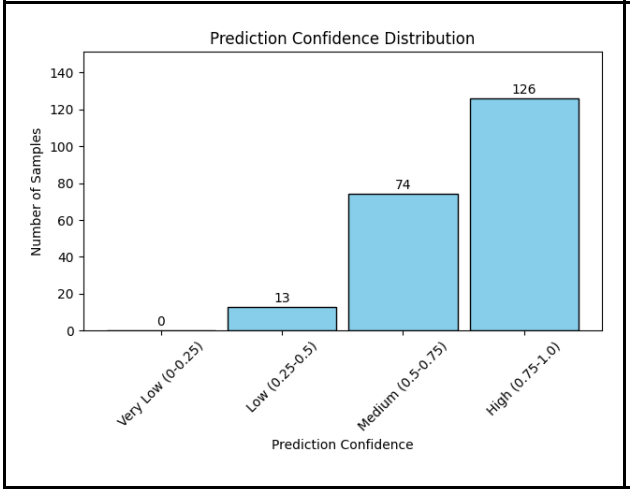

Model Testing 2 (9/3/26)

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--- Running Model Evaluation ---

--- Core performance metrics ---

[Test 1] Accuracy: 67.84%

[Test 2] Weighted F1-score: 67.22%

[Test 3] Classification Report:
              precision    recall  f1-score   support

  Background       0.68      0.53      0.60        43
     Comment       0.88      0.83      0.86        18
        Like       0.60      0.60      0.60        52
        Play       0.95      0.91      0.93        57
      Search       0.53      0.88      0.66        41
   Subscribe       0.55      0.36      0.44        44

    accuracy                           0.68       255
   macro avg       0.70      0.69      0.68       255
weighted avg       0.69      0.68      0.67       255

[Test 4] Confusion Matrix:
[[23  1  2  3 14  0]
 [ 0 15  2  0  1  0]
 [ 3  0 31  0  7 11]
 [ 2  1  0 52  0  2]
 [ 5  0  0  0 36  0]
 [ 1  0 17  0 10 16]]

[Test 5] ROC-AUC Score:
ROC-AUC Score (multi-class, OVR): 0.9235

[Test 6] Per-class accuracy:
Background: 53.49%
Comment: 83.33%
Like: 59.62%
Play: 91.23%
Search: 87.80%
Subscribe: 36.36%


--- Feature analysis ---

[Test 7] Top Feature Importances:
duration             0.254155
pk_count             0.202493
avg_inbound_size     0.160064
avg_outbound_size    0.148323
avg_iat              0.122425
std_iat              0.112540
dtype: float64
```
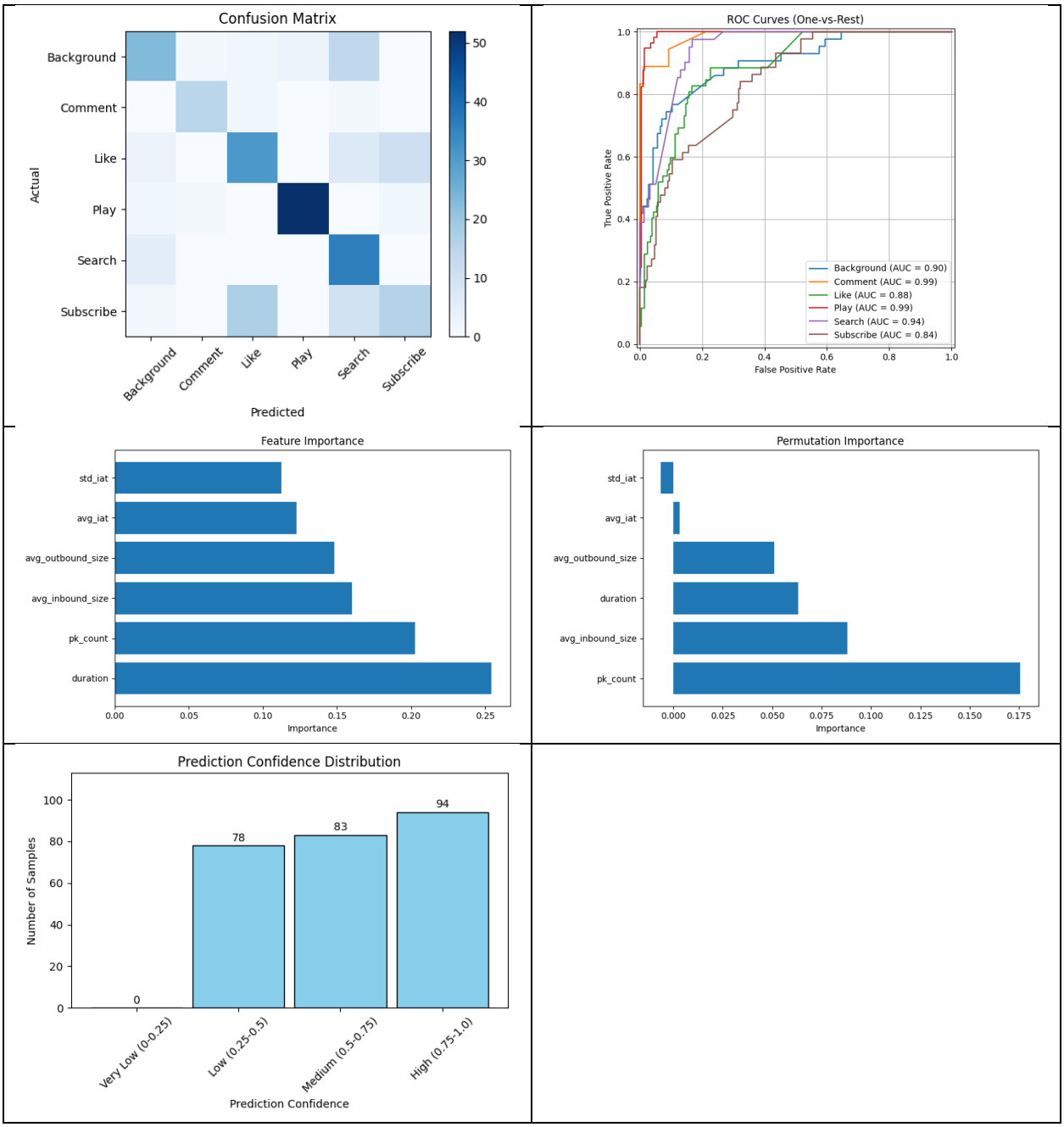

```
[Test 8] Permutation Importance:
pk_count             0.175686
avg_inbound_size     0.088235
duration             0.063137
avg_outbound_size    0.050980
avg_iat              0.003137
std_iat             -0.006275
dtype: float64

[Test 9] Feature ablation results:
Baseline accuracy: 67.84%
Without duration: 67.06%
Without std_iat: 69.02%
Without avg_iat: 67.06%
Without pk_count: 65.88%
Without avg_inbound_size: 67.84%
Without avg_outbound_size: 67.84%


--- Model robustness & reliability ---

[Test 10] Overfitting check:

Training Accuracy: 77.38%
Testing Accuracy:  67.84%

[Test 11] Class distribution:
Total samples: 790
action
Like          150
Play          150
Subscribe     150
Background    150
Search        138
Comment        52
Name: count, dtype: int64

[Test 12]Prediction confidence (first 10 rows):
Sample 0: 0.676
Sample 1: 0.667
Sample 2: 0.530
Sample 3: 0.779
Sample 4: 0.450
Sample 5: 0.475
Sample 6: 0.688
Sample 7: 0.288
Sample 8: 0.450
Sample 9: 0.283

[Test 13] Noise sensitivity test:
Noise 0.5% -> Accuracy: 68.24%
Noise 1.0% -> Accuracy: 68.24%
Noise 5.0% -> Accuracy: 68.63%
```

```
[Test 14] Stability test:

Stability scores: [0.7241379310344828, 0.7164750957854407, 0.7279693486590039, 0.7394636015325671, 0.7126436781609196]
Mean stability: 0.7241379310344828


--- Error & unseen label analysis ---

[Test 15] Error analysis:

Misclassified samples:
      duration   std_iat   avg_iat  ...  avg_outbound_size  true        pred
202   1.402640  0.122197  0.035066  ...         128.375000  Like  Background
265   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
489   2.632032  0.138046  0.039879  ...         775.848485  Like   Subscribe
630   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
764   4.100947  0.188961  0.048246  ...         705.047059  Like   Subscribe
860   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
892   0.013041  0.004767  0.002608  ...         110.400000  Like      Search
1154  1.495321  0.583267  0.249220  ...         145.000000  Like  Background
1338  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
1432  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
1544  2.278423  0.126649  0.052987  ...        2656.372093  Like   Subscribe
1594  2.332916  0.126422  0.044017  ...        1927.886792  Like   Subscribe
1653  2.297378  0.103865  0.028717  ...         679.475000  Like   Subscribe
1743  2.985846  0.122321  0.040902  ...         782.232877  Like   Subscribe
1934  1.617230  0.129754  0.041467  ...         118.179487  Like  Background
2114  2.841430  0.099818  0.033040  ...         710.290698  Like   Subscribe
2221  3.127080  0.102260  0.039089  ...         720.462500  Like   Subscribe
2267  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
2322  3.250708  0.186127  0.056047  ...        2236.655172  Like   Subscribe
2354  2.341362  0.236726  0.055747  ...        2138.880952  Like   Subscribe

[20 rows x 8 columns]
Total samples: 255
Total misclassified: 82
```

```
[Test 16] Testing model handling of unseen labels:

Unknown label handling test passed.
Model correctly raises error for unseen class:
Error: y contains previously unseen labels: 'new_action'

[test 17] General Sanity check (first 5 rows of features):
     duration   std_iat   avg_iat  pk_count  avg_inbound_size  avg_outbound_size
94   1.918529  0.116340  0.042634        45               0.0        2373.488889
163  1.970391  0.118672  0.042835        46               0.0        2303.956522
202  1.402640  0.122197  0.035066        40               0.0         128.375000
239  2.449518  0.116469  0.035500        69               0.0         682.898551
265  0.000000  0.000000  0.000000         1               0.0          66.000000

--- Testing complete ---
```



Model Performance on User Uploaded PCAP (Changes + Tests)